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# Access-Induced Semantic Divergence in Generated Program Transformations

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## Abstract

1 Access-induced semantic divergence occurs when an operation that appears obser-  
2 vational updates latent access state and changes a later externally visible result. We  
3 model this pattern as an observation-sensitive deterministic system (OSDS), prove  
4 structural properties for a restricted straight-line core, and use the model to build  
5 real-package witnesses and OSDS-aware replay checks. The real-code study con-  
6 firms 20 divergences across 12 unmodified Python packages, with 9 caller-level  
7 branch effects and 19/19 applicable mechanism controls eliminating divergence.  
8 In a frozen generated-transformation benchmark, five model-task outputs over  
9 two underlying package witnesses produced verified OSDS divergence. Ordinary  
10 benchmark checks missed all five; OSDS-aware verification caught all five; exact  
11 replay reproduced all five; and mechanism-neutralizing controls removed all five  
12 while the generated candidate stayed fixed. A prospectively frozen expansion from  
13 7 additional confirmed witnesses produced 42 outputs: 23/42 were task-compliant  
14 transformations, 19/42 returned unchanged code, 23/23 compliant transformations  
15 preserved OSDS behavior, and 0/42 produced verified OSDS divergence. The re-  
16 sults support a bounded conclusion: generated transformations can silently vio-  
17 late access-sensitive semantics on validated witnesses, and a mechanism-specific  
18 OSDS model with replay and causal controls provides a verifiable method for de-  
19 tecting and attributing this failure mode.

## 20 1 Introduction

21 Program transformations often treat logging, representation, inspection, caching, and repeated ac-  
22 cess as behavior-preserving. That treatment is valid when the accessed operation is observational  
23 in the semantic sense. It can fail when the operation updates latent state that a later read exposes.  
24 We call this pattern access-induced semantic divergence. OSDS adds structure beyond detecting  
25 unequal executions: it models the latent state changed by observation-shaped accesses, defines the  
26 access-order witness to compare, and specifies neutralizing interventions for attribution.

27 The phenomenon is related to observer effects and probe effects, but the paper studies a specific  
28 sequential mechanism: an observation-shaped access mutates latent access state, and a later program  
29 result changes. This matters for generated code because requested edits are often local and plausible.  
30 A model may add instrumentation, memoize a representation, or simplify repeated access while  
31 preserving ordinary tests that do not vary observation order.

32 We study this mechanism through three evidence levels. First, phenomenon breadth: the real-code  
33 oracle confirms 20 divergences across 12 unmodified Python packages, with 9 caller-level effects  
34 and 19/19 applicable real-code mechanism controls eliminating divergence. Second, generated-code  
35 failures: the frozen primary benchmark contains five verified model-task OSDS failures over two  
36 underlying package witnesses, pytest and PyYAML, and all five were silent under the benchmark or-

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Original shape:
  with catching_logs(handler, level):
    result = subject(logger, handler)

Generated instrumentation shape:
  logger.debug("diagnostic")
  with catching_logs(handler, level):
    result = subject(logger, handler)

State flow:
  inserted diagnostic access -> logging/handler state changes
  -> later catching_logs observes changed state
  -> caller-visible result differs

Verification:
  ordinary check PASS | OSDS preservation FAIL | neutralized replay PASS

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Figure 1: Compact pytest failure schema. The inserted log call looks like observation. In the verified Terra and Luna rows, it mutates logging/handler state that is read later. The same generated candidates pass the ordinary benchmark check, fail OSDS preservation, and pass after the diagnostic logger is isolated from the captured handler hierarchy.

37 dinary checks. Third, prospective limitation: a prospectively frozen expansion built from previously  
 38 confirmed unused witnesses produced 42 model outputs, of which 23 were task-compliant transfor-  
 39 mations, 19 returned unchanged code, 23/23 compliant transformations were OSDS-preserving, and  
 40 0/42 produced verified OSDS divergence.

41 The empirical claim is bounded. We do not estimate ecosystem prevalence, and the three tested  
 42 Codex task-model configurations are not independent model providers. The strongest generated-  
 43 code evidence is the causal chain in the primary benchmark: five generated transformations passed  
 44 ordinary checks, failed OSDS-aware verification, reproduced under exact replay, and disappeared  
 45 under mechanism-neutralizing controls while the generated candidate stayed fixed. These are five  
 46 model-task outcomes over two underlying package witnesses, not a frequency estimate.

47 **Contributions.** We make four contributions. First, we give an OSDS mechanism model and  
 48 structural properties for access-state-carrying sequential systems. Second, we give a real-code wit-  
 49 ness methodology and report 20 confirmed divergences across 12 packages, including 9 caller-level  
 50 branch effects. Third, we evaluate frozen generated transformations and identify five silent model-  
 51 task OSDS failures over two package witnesses. Fourth, we combine mechanism-neutralizing causal  
 52 attribution with a prospectively frozen null expansion that separates task compliance from semantic  
 53 preservation.

## 54 2 Motivating Real-Package Example

55 Figure 1 shows the shortest verified failure family from the primary study. The task asked for  
 56 behavior-preserving instrumentation around a pytest logging witness. The generated edit inserted  
 57 diagnostic logging that looked observational. In the witness, that access interacted with the captured  
 58 handler hierarchy used by `catching_logs`; the later handler-level behavior changed.

59 The example gives the intuition before the formal notation. The generated transformation did not  
 60 crash, did not fail the ordinary smoke check, and did not look like a destructive update at the edit  
 61 site. The divergence appeared only when the evaluation compared the later visible result across the  
 62 access order that the witness exposes. In the causal-control replay, the exact generated files were  
 63 reused. The only change was the witness environment: diagnostic logging was isolated from the  
 64 handler hierarchy responsible for the latent mutation. Under that neutralized mechanism, the OSDS  
 65 divergence disappeared.

66 This is a generated transformation failure, not a defect claim about pytest. It shows that a preserva-  
 67 tion checker for generated edits needs to model the latent state that observation-shaped accesses can  
 68 affect.

| Stage      | Baseline path      | Transformed path          | OSDS question                 |
|------------|--------------------|---------------------------|-------------------------------|
| Initialize | matched object     | matched object            | same visible setup            |
| Observe    | no extra access    | observation-shaped access | can latent state drift?       |
| Read       | later visible read | later visible read        | does exposed value change?    |
| Classify   | ordinary check     | OSDS comparison           | is divergence access-induced? |

Figure 2: OSDS evaluation compares access-order variants that ordinary single-order checks may not exercise.

### 3 Observation-Sensitive Deterministic Systems

An OSDS configuration contains a semantic value  $x = (b, a, d)$  and an accumulator  $y$ . The component  $b$  is stable,  $a$  is an access count, and  $d$  is latent drift. A read exposes a value and updates access state:

$$\text{read}(b, a, d, y) = ((b, a + 1, r(a, d)), y + f(b, a, d)).$$

An observation exposes no additive value but can update latent drift:

$$\text{obs}(b, a, d, y) = ((b, a, g(d)), y).$$

A program body is a deterministic fold over a finite operation sequence  $\pi \in \{\text{read}, \text{obs}\}^*$ . A cap  $C$  maps the final body configuration to the externally visible result.

For a fixed initial configuration  $\text{cfg}_0$ , body sequence  $\pi$ , and cap  $C$ , let  $\llbracket \pi \rrbracket_C(\text{cfg}_0)$  denote the final visible output. Given a multiset of operations  $M$  and an admissible ordering set  $\Pi(M)$ , semantic divergence is

$$\Delta_C(M, \text{cfg}_0) = \max_{\pi \in \Pi(M)} \llbracket \pi \rrbracket_C(\text{cfg}_0) - \min_{\pi \in \Pi(M)} \llbracket \pi \rrbracket_C(\text{cfg}_0).$$

A transformation is OSDS-preserving for a witness when the baseline and candidate agree under the witness’s OSDS relation. In the benchmark, this relation is implemented as a package-specific metamorphic oracle and then manually reviewed for access-induced attribution.

### 4 Structural Results

The structural core is a straight-line deterministic template that supports the witness construction and causal-control interpretation. Within that scope, fixed-order execution is deterministic: induction over the operation sequence gives a unique next configuration at each step. Two zero-divergence conditions follow directly. If  $g(d) = d$ , observations are inert and can be moved without changing the reduced read sequence. If  $f(b, a, d) = h(b)$ , reads ignore access count and drift, so every ordering with the same number of reads produces the same body accumulator.

A linear cap preserves body-level divergence. For  $C(y) = \alpha y + \beta$  with  $\alpha \neq 0$ ,  $y_1 \neq y_2$  implies  $C(y_1) - C(y_2) = \alpha(y_1 - y_2) \neq 0$ . This proposition is the only cap result used as an unbounded structural claim. Nonlinear amplification and broad degree laws are treated as bounded computational findings.

The appendix states a restricted adjacent-swap result for the monotone positive-parameter fragment. Under deterministic updates, monotone positive observation drift, and monotone access-sensitive reads, extrema over the checked order set occur at boundary orderings obtained by adjacent swaps that move observations before or after reads. The appendix also states the restricted range-degree result supported by exact rational checks: for checked compositional OSDS cases  $m = 1..4$ , cap degree  $d = 1..5$ , and  $L = 2..15$ , extremal-order interpolation detects degree  $2d$ . This is not a general theorem for all OSDS programs.

### 5 Real-Code Study

The real-code study uses static screening only as a review queue. The screen covered 73 PyPI packages and 278 reviewed MEDIUM/HIGH findings. Manual review labeled 203 likely true positives and 75 likely false positives, for reviewed precision 0.7302. A rebuilt exact-version source snapshot

| Real-code evidence item                             | Count |
|-----------------------------------------------------|-------|
| Reviewed static findings                            | 278   |
| Executable harnesses                                | 39    |
| Confirmed divergences                               | 20    |
| Confirmed packages                                  | 12    |
| Caller-level wrappers changing downstream execution | 9     |
| Mechanism controls eliminating divergence           | 19/19 |

Table 1: Real-code OSDS evidence. Static screening is a review queue; executable harnesses and controls provide the semantic evidence.

| Model         | Exec. | Pres. | Ord. bug | Invalid | OSDS fail | Silent OSDS |
|---------------|-------|-------|----------|---------|-----------|-------------|
| gpt-5.6-sol   | 26    | 23    | 3        | 0       | 0         | 0           |
| gpt-5.6-terra | 24    | 18    | 4        | 2       | 2         | 2           |
| gpt-5.6-luna  | 24    | 17    | 4        | 2       | 3         | 3           |

Table 2: Primary frozen coding-agent results. Exec. is executable outputs; Pres. means behavior preserved under the benchmark evaluation, not merely executable; Ord. bug means ordinary programming bug. Silent OSDS means a manually verified OSDS divergence for which the benchmark ordinary check passed.

104 contained 4,383 analyzable classes, and a deterministic sample of 200 unflagged classes found 0  
105 likely missed matches. These numbers are audit evidence rather than prevalence evidence.

106 Executable confirmation is the decisive filter. The oracle selected 60 candidates, constructed 39  
107 executable package-shaped harnesses, and confirmed 20 divergences across 12 unmodified pack-  
108 ages: httpcore, PyYAML, pytest, markdown, more-itertools, docutils, beautifulsoup4, boltons, cer-  
109 berus, dnspython, h11, and anyio. The divergences include stream materialization, identity-cache  
110 reuse, handler-level mutation, reference-registry mutation, cursor advance, destructive tree extrac-  
111 tion, cache recency/statistics, validation-error population, and buffer consumption.

112 Nine confirmed divergences were lifted into caller-level wrappers that changed branch labels and  
113 downstream consequences. Nineteen controls eliminated divergence under fresh-object, reset-  
114 between, or pure-observation interventions. These controls show that the detected differences de-  
115 pend on the access-induced mechanism named by the witness.

## 116 6 Generated Transformation Study

117 The primary benchmark was frozen before model execution. It contains 13 base tasks and 26 nor-  
118 mal/warned variants from 9 packages and 9 unique witnesses. The six transformation families are  
119 instrumentation, caching/materialization, refactoring, repeated-access cleanup, access reordering,  
120 and debugging/inspection. Each generation task was fresh and projectless. The model saw only  
121 the frozen prompt, with no OSDS labels, oracle output, witness labels, benchmark explanation, or  
122 previous responses.

123 Replay treats generated text as data. Raw responses are saved exactly, code is extracted, ordinary  
124 smoke checks are run, and then OSDS-aware checks compare baseline and candidate behavior under  
125 the witness relation. Manual review classifies non-preserved rows as verified semantic divergence,  
126 ordinary programming bug, invalid patch, environment failure, oracle issue, or unclear.

127 The benchmark includes hidden-observation rows, where an access looks observational in the re-  
128 quested edit and reveals its effect only through later state, and expected-access-sensitive rows, where  
129 consumption or mutation should be visible from the code shape. The five verified OSDS failures are  
130 five model-task outcomes over two underlying package witnesses. Terra failed on two pytest instru-  
131 mentation rows. Luna failed on those two pytest rows and one PyYAML caching/materialization row.  
132 Sol preserved the verified-OSDS rows on which Terra or Luna diverged. Expected-access-sensitive  
133 rows produced ordinary bugs or invalid patches, with no verified OSDS divergence.

| Family                 | Intervention                                              | Rows | Result      |
|------------------------|-----------------------------------------------------------|------|-------------|
| pytest instrumentation | isolate diagnostic logger from captured handler hierarchy | 4    | 4/4 removed |
| PyYAML caching         | clear representer identity cache after observation        | 1    | 1/1 removed |

Table 3: Causal controls. Under the original witness all five failures reproduced; under mechanism-neutralizing controls all five OSDS divergences disappeared.

| Model         | Outputs | Task-compliant | Unchanged | Ordinary pass | Ver. OSDS |
|---------------|---------|----------------|-----------|---------------|-----------|
| gpt-5.6-sol   | 14      | 8              | 6         | 14            | 0         |
| gpt-5.6-terra | 14      | 8              | 6         | 14            | 0         |
| gpt-5.6-luna  | 14      | 7              | 7         | 14            | 0         |
| Total         | 42      | 23             | 19        | 42            | 0         |

Table 4: Prospective expansion with task compliance separated from semantic preservation. Unchanged outputs are OSDS-preserving but are not counted as task-compliant transformations.

134 Self-assessment was secondary. Across Sol, Terra, and Luna, there were zero false YES preservation  
135 claims on verified OSDS divergences. Conservative NO responses were common, so self-assessment  
136 should not replace execution replay.

## 137 7 Causal Attribution and Prospective Expansion

138 The causal-control experiment replays the exact generated files for the five verified failures. The  
139 candidate source remains unchanged. The intervention changes only the witness mechanism that  
140 caused access-induced divergence.

141 This supports the access-induced attribution. The failures are mechanism-dependent generated-code  
142 defects: ordinary checks pass, OSDS checks fail, and targeted neutralization removes divergence  
143 while preserving the generated candidate.

144 After the causal phase, 11 previously confirmed unused real-code witnesses were audited prospec-  
145 tively under stated eligibility criteria. Seven met those criteria and were frozen before model ex-  
146 ecution, producing 14 normal/warned variants from boltons, dnspython, and h11. Sol, Terra, and  
147 Luna were run on these exact prompts as fresh projectless tasks. Self-assessment was skipped for  
148 the expansion.

149 The prospective benchmark produced no new verified OSDS divergences. The corrected interpre-  
150 tation is narrower than a 42/42 transformation success claim: 23/42 outputs performed a nontrivial  
151 task-compliant transformation, 19/42 returned unchanged code, and 23/23 task-compliant outputs  
152 were OSDS-preserving. The result constrains failure incidence among these frozen responses, sup-  
153 ports reporting the primary failures as reproducible examples rather than as a prevalence estimate,  
154 and provides evidence against treating the selected witness construction as mechanically failure-  
155 producing.

## 156 8 Related Work

157 **Observer effects and Heisenbugs.** Heisenbugs and probe effects describe programs whose behav-  
158 ior changes under observation or debugging. Musuvathi et al. discuss debugging statements affect-  
159 ing concurrent interleavings in systematic testing [Musuvathi et al., 2008]. Sallinger, Weissenbacher,  
160 and Zuleger formalize Heisenbugs as hyperproperties and analyze causes including probe effects and  
161 nondeterminism [Sallinger et al., 2026]. OSDS is narrower: it models sequential access-state mecha-  
162 nisms where observation-shaped operations mutate latent state and later reads expose the difference.

163 **Semantic preservation and equivalence.** Compiler verification and translation validation study  
164 semantic preservation between source and transformed programs [Pnueli et al., 1998, Leroy, 2009].  
165 OSDS uses the same preservation motivation but focuses on generated source-level transformations

over library objects whose read and observation operations carry latent state. The oracle compares access-order behavior rather than proving full program equivalence.

**Metamorphic testing.** Metamorphic testing addresses oracle problems by checking relations across related executions [Chen et al., 1998, 2018]. The OSDS-aware oracle is a specialized metamorphic relation: the follow-up execution changes the placement of an observation-shaped access while keeping witness setup matched. Generic metamorphic testing supplies the testing pattern; OSDS supplies the mechanism and classification boundary.

**Current generated-transformation benchmarks.** Differential fuzzing has recently been applied to LLM-generated code refactorings, showing that generated refactors can be functionally nonequivalent and that existing tests can miss a substantial fraction of those cases [Dristi and Dwyer, 2026]. SWE-Refactor moves refactoring evaluation to repository-level Java changes mined from real projects, emphasizing realistic context and compound refactorings [Xu et al., 2026]. VERINA benchmarks verifiable code generation in Lean across code, specification, and proof generation tasks [Ye et al., 2025]. Viverra studies text-to-code with generated assertions and bounded verification, surfacing verified properties alongside generated C programs [Wu et al., 2026]. Metamorphic transformations have also been used to diagnose memorization in LLM-based program repair; we use that line only as related evidence that semantics-preserving transformations can support stronger code-evaluation protocols [Koning et al., 2026].

**OSDS mechanism attribution.** Generic differential and metamorphic testing can establish that two executions differ under a chosen relation. OSDS contributes a sequential semantic model for objects whose reads and observations interact through latent access state. That structure determines which access reorderings form meaningful witnesses and supports mechanism-specific controls. The causal interventions test whether a detected divergence disappears when the access-induced mechanism is neutralized while the generated candidate remains unchanged, giving mechanism attribution alongside behavioral detection.

**Generated code and repair verification.** HumanEval and Codex made execution-based evaluation central for code generation [Chen et al., 2021]. SWE-bench extends evaluation to issue-resolution patches in real repositories [Jimenez et al., 2024]. Automated program repair has long shown that patches can overfit tests [Le Goues et al., 2012, Smith et al., 2015]. This paper studies a different preservation failure: a generated transformation can pass the ordinary benchmark tests while violating an access-state relation that those tests do not exercise.

## 9 Limitations

The coding-agent benchmark is small and correlated by witness and package. Counts are benchmark outcomes, not prevalence estimates. The five verified primary failures are five model-task failures over two underlying package witnesses. The prospective expansion added seven witnesses and found zero new verified OSDS divergences.

The three model labels are Codex task-model configurations reported by the Codex runtime. They were collected through fresh projectless Codex tasks rather than the OpenAI API, and temperature/seed were not exposed. The study is Python-focused. Manual review remains part of verified OSDS classification. The formal results apply to the stated straight-line deterministic core; broader analyzer and range-degree claims remain empirical or bounded computational evidence.

The intended positive impact is better preservation testing for generated code in libraries where observation-shaped accesses can mutate hidden state. The main risk is that the same examples could encourage superficial transformations that pass ordinary tests while changing access-sensitive behavior. We mitigate this risk by releasing bounded witnesses, replay scripts, and mechanism-neutralizing controls rather than a general attack system, and by framing the results as verification evidence rather than deployment guidance.

## 10 Conclusion

Access-induced semantic divergence is a concrete preservation risk for generated transformations on access-state-carrying software objects. The OSDS model identifies the mechanism, the real-code study confirms package witnesses, and the coding-agent benchmark shows five silent generated

failures under ordinary checks. Exact replay plus mechanism-neutralizing controls reproduced and removed all five primary failures. The prospective expansion found no new OSDS divergences and exposed substantial unchanged-output behavior. The resulting assessment is bounded: OSDS-aware verification provides a reproducible method for detecting and attributing a semantic failure mode that ordinary checks missed in this benchmark, without supporting broad claims of model unsafety.

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## 264 A Formal Details

### 265 A.1 State and transitions

266 An OSDS state is  $s = (b, a, d, y)$  with base  $b$ , access count  $a$ , latent drift  $d$ , and accumulator  $y$ . A  
267 read transition is deterministic:

$$R(s) = (b, a + 1, r(a, d), y + f(b, a, d)).$$

268 An observation transition is deterministic:

$$O(s) = (b, a, g(d), y).$$

269 For a finite sequence  $\pi = o_1 \dots o_n$ , composition is  $\llbracket \epsilon \rrbracket(s) = s$  and  $\llbracket \pi o \rrbracket(s) = o(\llbracket \pi \rrbracket(s))$ . Branch  
270 evaluation is a deterministic predicate  $B(C(\llbracket \pi \rrbracket(s)))$  over the capped visible result or final configu-  
271 ration. Branch divergence occurs when two admissible orderings produce different branch labels or  
272 downstream consequences.

### 273 A.2 Semantic divergence

274 Given operation multiset  $M$ , admissible orderings  $\Pi(M)$ , initial state  $s_0$ , and deterministic cap  $C$ ,  
275 define  $V_C(\pi, s_0) = C(\llbracket \pi \rrbracket(s_0))$ . The access-order range is  $\Delta_C(M, s_0) = \max_{\pi \in \Pi(M)} V_C(\pi, s_0) -$   
276  $\min_{\pi \in \Pi(M)} V_C(\pi, s_0)$  for finite  $\Pi(M)$ . A transformation preserves the witness when baseline and  
277 candidate have equal OSDS-observable results under the witness relation.

### 278 A.3 Fixed-order determinism

279 **Proposition.** For fixed  $s_0$ , fixed  $\pi$ , deterministic  $R$ , deterministic  $O$ , and deterministic  $C$ ,  $V_C(\pi, s_0)$   
280 is unique. **Proof.** Induct on  $|\pi|$ . The empty sequence returns  $s_0$ . For  $\pi o$ , the induction hypothesis  
281 gives a unique intermediate state and  $o \in \{R, O\}$  maps it to a unique next state. The deterministic  
282 cap gives a unique output.

### 283 A.4 Zero divergence under no observation-induced mutation

284 **Proposition.** If  $g(d) = d$  and observations expose no additive value, moving observations within a  
285 fixed multiset does not change the body accumulator or final latent state. **Proof.** Each observation  
286 is the identity on  $(b, a, d, y)$  and contributes no value to the accumulator. Delete observations from  
287 any admissible ordering; the remaining read sequence has the same length and order because all  
288 reads are identical. The same deterministic reads are therefore executed from the same state and  
289 contribute the same accumulator terms. Re-inserting identity observations at arbitrary positions  
290 leaves the accumulator and final latent state unchanged.

### 291 A.5 Zero divergence under access-insensitive reads

292 **Proposition.** If  $f(b, a, d) = h(b)$ , every ordering with  $L$  reads has the same body accumulator  
293  $Lh(b)$ . **Proof.** Every read contributes the same exposed value for fixed  $b$  regardless of access count  
294 or drift. Observations may change  $d$ , but  $d$  is ignored by  $f$ . A deterministic cap depending only on  
295 the accumulator and final counts preserves equality.

### 296 A.6 Linear-cap preservation

297 **Proposition.** Let  $C(y) = \alpha y + \beta$  with  $\alpha \neq 0$ . If body accumulators  $y_1 \neq y_2$ , then  $C(y_1) \neq C(y_2)$ .  
298 **Proof.**  $C(y_1) - C(y_2) = \alpha(y_1 - y_2)$ , which is nonzero under exact arithmetic.

### 299 A.7 Adjacent-swap extrema under monotone access drift

300 **Restricted lemma.** Consider the positive monotone fragment with identical reads, identical ob-  
301 servations, deterministic updates, observation update  $g$  satisfying  $g(d) \geq d$ , and read contribution  
302  $f(b, a, d)$  monotone nondecreasing in  $d$  for fixed  $b, a$ . Assume observations do not change  $a$  or  $y$ ,  
303 reads increase  $a$  by one, and the final cap is monotone in the accumulated read contribution. For any



adjacent pair  $RO$  in an otherwise fixed ordering, replacing it by  $OR$  cannot decrease the contribution of the swapped read or any later read. **Proof.** Let the common prefix before the pair produce state  $(b, a, d, y)$ . In the local order  $RO$ , the read contributes  $f(b, a, d)$  and the following observation changes drift to  $g(d)$ . In the local order  $OR$ , the observation first changes drift to  $g(d)$ , and the read contributes  $f(b, a, g(d))$ . Since  $g(d) \geq d$  and  $f$  is monotone in drift,  $f(b, a, g(d)) \geq f(b, a, d)$ ; the read count after the pair is the same in both orders. The suffix after the pair therefore starts with the same  $b$  and  $a$ , an accumulator no smaller in the  $OR$  run, and drift  $g(d)$  in both runs. Inducting over later operations gives no smaller drift-visible read contributions and no smaller final capped value. Repeated adjacent swaps can move each observation left across neighboring reads without decreasing value, so an ordering with all observations before all reads is a maximum among orderings reachable by these swaps. The reverse argument moves observations right across reads without increasing value, giving the all-reads-before-observations boundary as a minimum. **Scope.** The lemma is limited to this positive monotone straight-line fragment with identical operations and monotone caps. It is not claimed for arbitrary Python programs or nonmonotone observation effects.

## 318 A.8 Restricted 2d range-degree statement

**Bounded exact computational result.** For the compositional OSDS cap family implemented in the exact-rational validation scripts, with  $m = 1.4$ , cap degree  $d = 1.5$ , and  $L = 2.15$ , exhaustive checks on small feasible grids plus extremal-order interpolation detect access-order range degree  $2d$ . **Derivation.** In the checked family, the body-range term is  $\eta mL(L-1)/2$ . The compositional cap contributes  $d-1$  final read factors, each quadratic in  $L$ . The leading degree is therefore  $2+2(d-1) = 2d$ . Exact rational scripts verify the extremal formula against enumerated feasible orderings and holdout values on the stated bounded domain. **Scope.** This is a restricted exact computational result for the named family and grid. It is not a general theorem for all OSDS programs.

## 327 B Benchmark Details

Primary tasks are in `benchmark/tasks.jsonl` under the agent-behavior study tree. Primary prompt files are in that tree's `prompts/` directory. The pre-model manifest and raw Codex task-model responses are in `external_collection/` and `external_collection/responses/`.

Manual review rows are in the primary study analysis directory as the 20260813 manual-review JSONL. The matching cross-model summary is stored beside it as JSON and Markdown. Prospective tasks, prompts, and raw responses are under `benchmark_expansion/`. Exact prospective replay outputs are in the three `*expansion-exact-20260813Tcutscope` result directories.

The prospective task-compliance audit is `analysis/prospective_task_compliance.csv`. It separates `task_compliance`, `ordinary_correctness`, and `osds_preservation` for all 42 prospective outputs. The supplement artifact index maps these shortened descriptions to exact relative paths.

## 339 C Reproducibility

The reviewer supplement contains a relative-path artifact index, frozen prompts, raw response JSONLs, replay summaries, causal-control scripts, real-code oracle metadata, and expected outputs. It excludes virtual environments, caches, authentication material, and personal absolute paths. Replays require Python 3.11 or newer with `pytest` and the package versions named by the witness manifests. The supplement `REPRODUCE.md` gives exact commands for the primary replay, prospective replay, causal-control replay, and table reproduction.

All artifact replays are CPU-only Python executions and do not require a GPU, accelerator runtime, cloud worker, model credential, or model-provider API call. A pre-submission reproduction check on Windows 11 with Python 3.14.4, a 6-core/12-thread AMD Ryzen 5 4600H CPU, and 7.4 GiB RAM measured 1.12 s for a baseline single-task replay, 1.22 s for a frozen primary single-task replay, 1.13 s for a frozen prospective single-task replay, 3.49 s for the five-case causal-control replay, 0.57 s for the real-code witness validation command, and 0.13 s for the summary-table command. Exact peak memory was not logged; the commands operate on small JSONL, CSV, and Python files, and the packaged supplement is below 1 MB. The earlier Codex task-model generation phase is not

reproduced by the supplement. Its inference hardware, accelerator hours, and hidden service-side memory were not exposed by the Codex task interface, and preliminary or failed collection attempts used additional model calls outside the replay package.

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